
Determinants of Academic Success: A Socioeconomic and Demographic Analysis

Anil Brahmajosyula (NetID: ab2342)

Gordon Lin (NetID: gl548)

Rithwik Anand (NetID: ra989)

Abstract

This project will investigate the extent to which socioeconomic factors and demographic variables predict student academic performance in secondary education. Using the UCI Student Performance Dataset, we will analyze the impact of variables such as parental education, household income, and residential area on final grades. Our goal in mind is to identify the non-academic factors of academic success to inform more equitable educational policy and targeted intervention programs.

1 Research Question and Motivation

Research Question: To what extent do socioeconomic factors (parental education, household income) and demographic variables (age, residential area) predict a student's final academic performance in secondary education?

Motivation: Understanding the factors of academic success is important for developing equitable educational policies. By identifying which non-academic factors most heavily influence grades, educators can better implement targeted intervention programs for at-risk students. This research aims to move beyond simple correlation and towards causation, in order to identify the weight of institutional and environmental barriers in the educational landscape.

2 Data Sources

Data Identification This project will utilize the **Student Performance Dataset** from the UCI Machine Learning Repository, which contains student achievement data in secondary education from two Portuguese schools.

Obtainment and Preprocessing

- **Obtainment:** The data will be imported directly into a Python environment using the UCIMLRepo library or as a .csv file.
- **Cleaning:** We will handle binary variables (e.g., urban vs. rural) using one-hot encoding and normalize numerical variables like "absences" and "study time."
- **Challenges:** A potential challenge is the high dimensionality of socioeconomic features relative to the number of samples ($N = 649$). Another challenge may be the potential for data leakage if mid-term grades ($G1, G2$) are used to predict final grades ($G3$).

3 Methodology and Analysis Plan

Data Science Techniques: This project will apply supervised machine learning techniques to analyze the relationship between socioeconomic and demographic variables and student academic performance. In addition to predictive modeling, the study will apply feature importance analysis to determine which socioeconomic and demographic variables contribute most to predicting academic performance.

Approach

- **Exploration (EDA):** Exploratory data analysis can be used to understand the dataset using summary statistics and visualizations. This includes examining distributions of variables, detecting missing values and outliers, and using correlation matrices and plots to explore relationships between predictors (parental education, household income, absences) and the target variable (final grade). Additional exploratory techniques such as pair plots, boxplots, and distribution plots will be used to compare academic outcomes across demographic groups. Missing data patterns will also be examined to ensure data quality before modeling.

- **Modeling:** Build predictive models using regression techniques such as Linear Regression, Ridge/Lasso Regression, and Random Forest Regression. These models will be trained on the processed dataset to capture both linear and non-linear relationships between socioeconomic factors and academic performance.

Linear Regression will serve as a baseline model to estimate the direct linear relationship between socioeconomic predictors and student final grades. Ridge and Lasso Regression will be applied to address potential multicollinearity among predictors and to perform regularization, which helps prevent overfitting and improves model generalization. In particular, Lasso regression can also assist in feature selection by shrinking less important coefficients toward zero.

In addition, Random Forest Regression, an ensemble tree-based method, will be used to capture more complex and non-linear relationships between demographic or socioeconomic variables and academic performance. Random Forest models also provide measures of feature importance, which can help identify which variables contribute most strongly to predicting student outcomes. Comparing these models will allow the study to evaluate both interpretable linear approaches and more flexible machine learning methods.

- **Evaluation:** Split the dataset into training and testing sets and apply cross-validation to ensure robust model performance. Compare model predictions with actual values to evaluate accuracy and determine which model performs best.

Tools The analysis will be implemented using Python and common data science libraries including Pandas for data manipulation, NumPy for numerical operations, Matplotlib and Seaborn for visualization, and Scikit-Learn for machine learning models and evaluation. The analysis will be conducted in a Jupyter Notebook environment.

4 Expected Outcomes and Evaluation

Anticipated Findings The study is expected to reveal which socioeconomic and demographic factors have the strongest influence on student academic performance. Variables such as parental education, study time, and absences may show significant relationships with final grades. These findings could help identify key non-academic factors contributing to academic success or risk of poor performance.

Metrics

- **R^2 Score:** The R^2 score will be used to measure how well the model explains the variance in student final grades. A higher R^2 value indicates that the model better captures the relationship between the predictors and the target variable.

Broader Implications The findings may provide insights for educators and policymakers about how socioeconomic conditions affect student outcomes. Understanding which socioeconomic and demographic factors influence academic performance can help educators and policymakers support struggling students and reduce educational inequality.